

Lesson 02 – Chunking (Interview Notes)

1. Why does chunking exist?

Chunking divides large documents into smaller, manageable pieces before they are sent to the LLM. This is necessary because sending an entire document for every query can exceed the model's context window, increase inference latency, increase API cost, and reduce answer quality by introducing large amounts of irrelevant information.

2. Why can't we send the whole document to the LLM?

There are several reasons:

- **Context Window Limitation:** Large documents may exceed the maximum number of tokens the model can process.
 - **Latency:** More input tokens require more computation, resulting in slower responses.
 - **Cost:** For API-based models, more input tokens directly increase the cost of each request.
 - **Accuracy:** Large amounts of irrelevant context can distract the model from the information needed to answer the user's question.
 - **Scalability:** In production systems with thousands or millions of documents, sending the entire knowledge base for every query is not feasible.
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3. What is a Context Window?

A context window is the maximum amount of information an LLM can process during a single request. It typically includes:

- System Prompt
- Conversation History
- User Prompt
- Retrieved Context (RAG documents)
- Model Response (generated output)

Some reasoning models also internally use reasoning tokens, depending on the model and provider.

4. Why is overlap needed?

Overlap preserves context between consecutive chunks. Without overlap, important information such as sentences, paragraphs, or concepts may be split across chunk boundaries, causing the model to miss relevant context during retrieval. Overlap acts as a bridge between neighboring chunks and improves retrieval quality.

5. Step Formula

The distance moved before creating the next chunk is:

```
step = chunk_size - overlap
```

For example:

- Chunk Size = 500
- Overlap = 100
- Step = 400

This means each new chunk starts 400 characters (or tokens, depending on the implementation) after the previous one, while retaining the last 100 characters as shared context.

Interview Tip

Question: Why do we use overlap?

Answer: We use overlap to preserve context across chunk boundaries. If a sentence or important concept is split between two chunks, overlap ensures that the complete information appears in at least one retrieved chunk, improving retrieval accuracy.